

Olympiad Test Paper — Social Science (Class 7) — Paper 3

Chapter 19: Infrastructure: Engine of India's Development

Paper 3 — (progressively harder)

Subject	Social Science (NCERT Class 7)
Chapter	19 — Infrastructure: Engine of India's Development
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Classic traps: misclassifying invisible utilities or social services as 'not infrastructure'; assuming the British built the 1853 railways to help Indians rather than to serve their own extraction and control; treating infrastructure care as the government's job alone; confusing physical sub-types (transport vs utilities vs communication vs energy) with one another and with social infrastructure; mistaking renewable energy parks, living root bridges, or the Arthashastra's concern as outside the chapter's scope. Do not write on this paper — mark answers on your answer sheet.

1. A planner lists four things her district needs: a new fibre-optic network, a district court, a water pipeline, and a wind farm. She wants to know how many of these are PHYSICAL infrastructure (not social). The correct count is:

- (A) Two
- (B) One
- (C) Three
- (D) All four

2. The Latin roots of 'infrastructure' are infra and structura. Using only the meanings of these roots, the BEST literal reading of the word is:

- (A) The tallest structure standing above a city
- (B) A structure built quickly and cheaply
- (C) The underlying structure that lies beneath and supports everything built on top
- (D) A structure shared between far-apart places

3. Someone argues: 'Anything you cannot see cannot be infrastructure.' The strongest single example from the chapter that REFUTES this is:

- (A) A multi-storey hospital building, which everyone can see
- (B) A national highway, which is clearly visible

- (C) Underground power grids and water pipelines, which are essential yet largely hidden from view
- (D) A sea port crowded with ships, plainly in view

4. Infrastructure is split into exactly two main types. A student proposes the pair 'expensive and cheap'. The reason this is the WRONG basis for the chapter's division is that:

- (A) The chapter divides infrastructure by FUNCTION into physical and social, not by cost
- (B) Infrastructure actually has three official types, not two
- (C) All infrastructure costs the same, so cost cannot divide it
- (D) Cheap infrastructure is never counted as infrastructure

5. Which statement combines two ideas from the chapter CORRECTLY?

- (A) Infrastructure is the engine of development, and it spans both physical systems and social services
- (B) Infrastructure is the engine of development, but it covers only roads and buildings
- (C) Infrastructure is the final result of development, and it includes only social services
- (D) Infrastructure is the enemy of development, and it includes only invisible systems

6. Match each facility to its correct type: (i) a sewage-treatment and sanitation plant, (ii) a children's public park, (iii) a 4G mobile-tower network. The correct classification set is:

- (A) (i) social, (ii) social, (iii) social
- (B) (i) physical, (ii) physical, (iii) social
- (C) (i) physical utility, (ii) social, (iii) physical communication
- (D) (i) social, (ii) physical, (iii) physical

7. A town has excellent roads, power and ports but almost no schools, hospitals or sanitation. According to the chapter's logic, this town is most likely to:

- (A) Develop perfectly, since physical infrastructure alone guarantees progress
- (B) Need to demolish its roads and ports to make room for development
- (C) Be unaffected, because social infrastructure plays no real role in development
- (D) Struggle to develop fully, because development needs healthy, educated people as much as good physical systems

8. Why does the chapter insist that a hospital and a power grid, though very different, both belong under 'infrastructure'?

- (A) Both are physical infrastructure because both use electricity
- (B) Both are social infrastructure because both serve people
- (C) Neither is infrastructure; only roads are infrastructure
- (D) Both are basic facilities a country needs to function — the hospital as social, the grid as physical infrastructure

9. A facility provides electricity to homes, factories, schools and hospitals alike. Its dual significance, per the chapter, is that it is:

- (A) Social infrastructure, because it serves people directly
- (B) Communication infrastructure, because it sends signals to homes
- (C) Physical (energy/utility) infrastructure that other infrastructure depends on to operate
- (D) Not infrastructure, because households also use it privately

10. Of the following, the one that is NOT physical infrastructure is:

- (A) A natural-gas pipeline
- (B) A telecom satellite uplink station
- (C) A district court complex
- (D) A hydroelectric dam

11. Two regions, X and Y, are compared. X has reliable power, good highways and schools; Y has none of these. The chapter's reasoning predicts that, other things equal:

- (A) Y will develop faster because it has less to maintain
- (B) X is positioned to develop more readily because both its physical and social infrastructure support economic activity and human wellbeing
- (C) Both will develop identically, since infrastructure does not affect development
- (D) X will fall behind because too much infrastructure slows a region down

12. The chapter calls infrastructure the 'backbone' of the economy. The most precise meaning of this metaphor is that:

- (A) Infrastructure is the single most expensive thing an economy owns
- (B) Like a backbone supporting a body, infrastructure carries and connects the economic activities that depend on transport, power, water and communication
- (C) Infrastructure is rigid and cannot ever be changed once built
- (D) Infrastructure is only useful to the body's largest organs, not small ones

13. A factory owner says good infrastructure cut his costs. The chain of reasoning that best fits the chapter is:

- (A) Infrastructure raised his costs because maintaining roads is expensive for factories
- (B) Infrastructure helped only by reducing the number of workers he needed
- (C) Better roads and ports speed the movement of raw materials and finished goods, and reliable power keeps machines running, so production is cheaper and faster
- (D) Infrastructure made no difference; only the price of raw materials mattered

14. Which statement correctly distinguishes how PHYSICAL and SOCIAL infrastructure each aid development?

- (A) Physical infrastructure produces educated people; social infrastructure moves goods and power
- (B) Both kinds only move goods; neither affects people

(C) Physical infrastructure matters in cities only; social infrastructure matters in villages only

(D) Physical infrastructure moves goods, power and information; social infrastructure produces healthy, educated, capable people

15. India's road network is described as the world's second-largest. A student concludes 'so India must also have the world's second-largest railway passenger load.' The correct response is:

(A) Yes, road size and railway passenger rank are always the same

(B) No, because India has no railways at all

(C) No, because India's roads are actually the world's smallest

(D) The two facts are separate; the chapter states the road network is the world's second-largest and that the railways carry about 20 million people a day, but does not equate the two rankings

16. An expressway and an ordinary village lane differ in several chapter-relevant ways. The expressway is best described as:

(A) A slow local road maintained by a single village

(B) A high-speed national road built by the central government to link cities across states

(C) An elevated urban rail line within one city

(D) A coastal route used by cargo ships between ports

17. The Golden Quadrilateral links Delhi, Mumbai, Chennai and Kolkata. Its shape is called a 'quadrilateral' because:

(A) It is built only in four villages (B) It has exactly four lanes throughout

(C) It connects four foreign capital cities

(D) It connects four major cities, forming a four-sided road circuit across the country

18. NH44 is highlighted in the chapter. Which description best captures why it is significant among India's roads?

(A) It is one of India's major national highways, notable as among the longest, knitting distant parts of the country together

(B) It is a single city's metro line carrying daily commuters

(C) It is a sea port handling the most containers in India

(D) It is a solar park supplying the national grid

19. A goods company moves cargo from Delhi to Chennai. Using the chapter's framing, the Golden Quadrilateral helps it MOST by:

(A) Supplying electricity to the company's warehouses along the way

(B) Treating the company's sewage at each city

- (C) Providing a fast, direct highway link between major cities so goods reach distant markets quickly
- (D) Granting the company a patent on its product

20. Highways, railways, metros, airports and ports are all listed in the chapter. The single category that correctly contains ALL of them is:

- (A) Social infrastructure
- (B) Physical transport infrastructure
- (C) Communication infrastructure
- (D) Energy infrastructure

21. The chapter says the British began India's railways in 1853. A student writes, 'so the railways have always been India's lifeline since 1853.' The most accurate correction is:

- (A) Correct — the railways served Indian travellers from the very first day in 1853
- (B) The railways were never a lifeline and still serve only the British
- (C) The 1853 railways were first built for British colonial ends and only LATER became India's national lifeline
- (D) The railways were built in 1853 by Indians for free public travel

22. The British colonial motives for building the 1853 railways included all of the following EXCEPT:

- (A) Carrying India's raw materials to ports for export
- (B) Distributing British-made goods across India for sale
- (C) Moving troops quickly to keep control over the territory
- (D) Giving Indian villagers free, fast travel for their own benefit

23. The railways carry about 20 million people each day. A student misreads this as '20 people a day'. The scale he has misjudged is by a factor of about:

- (A) One million
- (B) One hundred
- (C) One thousand
- (D) Ten

24. The chapter uses the railways to illustrate a general principle about infrastructure. That principle is best stated as:

- (A) Infrastructure built by colonisers must always be torn down
- (B) Infrastructure can only ever serve the single purpose it was first built for
- (C) Infrastructure built long ago becomes useless within a few years
- (D) Infrastructure built for one purpose can later be repurposed to serve a very different, often wider, need

25. Which pair correctly contrasts a metro with the Indian Railways' long-distance network?

- (A) A metro links distant states; the Railways serve only one city
- (B) A metro is a fast urban rail system easing city congestion; the Railways form a nationwide network linking distant regions

(C) Both are sea-transport systems for cargo ships

(D) A metro is energy infrastructure; the Railways are communication infrastructure

26. Both the metro and a long-distance train are rail systems, yet the chapter treats the metro's role differently. The metro's distinctive contribution is:

(A) Moving large numbers of city commuters and reducing road traffic congestion

(B) Exporting raw materials to foreign ports

(C) Generating renewable electricity for the city

(D) Carrying cargo containers between coasts

27. The chapter notes aviation comes from the Latin avis ('bird'). A consistent extension of this reasoning is that aviation infrastructure centres on:

(A) Airports and aircraft that carry people and goods through the air

(B) Harbours and ships that carry cargo by sea

(C) Power stations that generate electricity (D) Pipelines that carry water to homes

28. India is said to have about 159 airports. If a student claims 'India therefore has 159 sea ports too,' the correct response is:

(A) Yes, airports and sea ports are always equal in number

(B) No — 159 is the chapter's airport figure; sea ports are a separate kind of facility on India's coasts and are not given that same number

(C) No, because India has no airports at all

(D) No, because all 159 are actually sea ports, not airports

29. Both an airport and a sea port move goods and people, but they belong to different transport modes. The shared category they both fall under is:

(A) Social infrastructure

(B) Energy infrastructure

(C) Communication infrastructure

(D) Physical transport infrastructure

30. Port A handles 5,000 TEUs in a day and Port B handles 8,000 TEUs the same day. Using the meaning of TEU, the correct interpretation is:

(A) Port B handled 3,000 more standard twenty-foot-equivalent containers that day, indicating greater container-handling capacity

(B) Port B ran its trains 3,000 km farther that day

(C) Port A is 3,000 metres taller than Port B

(D) Port B flew 3,000 more aircraft than Port A

31. Why is a standard unit like the TEU useful for comparing the world's ports?

(A) It gives a common container measure so ports of different sizes and cargo can be compared fairly

(B) It measures how fast metros run inside a port city

(C) It records the exact height of a port's tallest crane

(D) It counts how many airports a port owns

32. India's major sea ports lie on both its coasts. The geographic reason this matters for trade is that:

- (A) Ports on both the eastern and western coasts give India sea access in two directions for importing and exporting goods
- (B) Coastal ports are useless because ships cannot reach the interior
- (C) Ports must be in the Himalayas to handle the heaviest cargo
- (D) Ports on both coasts double the country's number of airports

33. A logistics manager plans to ship containers overseas and also fly urgent parcels abroad. The two pieces of physical infrastructure she relies on are, respectively:

- (A) A power grid for the containers and a metro for the parcels
- (B) A school for the containers and a hospital for the parcels
- (C) A sea port for the containers and an airport for the urgent parcels
- (D) A water pipeline for the containers and a court for the parcels

34. Someone claims 'all modern communication technology, including radio, was invented entirely in the West.' Drawing on the chapter, the strongest counter is:

- (A) The Indian scientist J.C. Bose demonstrated radio-wave communication and improved the 'coherer' detector, showing India contributed to the science of communication
- (B) India only began studying radio after independence in 1947
- (C) Radio waves were first used in ancient India for trade
- (D) The British invented radio when they built the railways in 1853

35. J.C. Bose's wireless work dates to about 1895. Placing this against other dates in the chapter, which ordering is chronologically correct (earliest first)?

- (A) J.C. Bose's wireless work (1895) → Indian Railways (1853) → independence (1947)
- (B) Independence (1947) → Indian Railways (1853) → Bose's wireless work (1895)
- (C) Indian Railways (1853) → independence (1947) → Bose's wireless work (1895)
- (D) Indian Railways (1853) → J.C. Bose's wireless work (1895) → Indian independence (1947)

36. A 'coherer', improved by J.C. Bose, played which role in early wireless?

- (A) It detected and decoded the transmitted wireless (radio) signals at the receiving end
- (B) It generated the electricity that powered the railways
- (C) It pumped clean water into city homes
- (D) It carried passengers along an elevated city track

37. A patent gives an inventor a temporary exclusive right. The chapter notes J.C. Bose 'chose not to patent much of his work.' The most reasonable inference about Bose's attitude is that he:

- (A) Believed his inventions did not work and were worthless
- (B) Was legally forbidden by the British from ever holding a patent

- (C) Wanted to keep all his findings secret from everyone forever
- (D) Was willing to let his scientific ideas be freely used rather than seek personal exclusive profit from them

38. If Bose HAD patented every invention, the chapter's definition of a patent tells us the most direct consequence would have been:

- (A) His inventions would automatically have stopped working
- (B) The British would have been forced to leave India sooner
- (C) Everyone else could have copied his inventions freely with no limit
- (D) For a number of years, only he (or those he permitted) could legally make, sell or use those inventions

39. Telecommunication comes from tele ('far') and communicare ('to share'). Which everyday activity is therefore an example of telecommunication, by this root meaning?

- (A) Driving a car along a national highway
- (B) Speaking to a relative in another city over the phone
- (C) Loading containers onto a ship at a port
- (D) Generating power at a wind farm

40. Communication infrastructure and transport infrastructure both 'connect' places, yet differ. The clearest distinction the chapter draws is that:

- (A) Communication moves goods; transport moves only information
- (B) Communication infrastructure moves information across distance, while transport infrastructure moves people and goods
- (C) Both move only electricity and nothing else
- (D) Communication works only abroad; transport works only in India

41. Among J.C. Bose's contributions noted in the chapter, the one correctly described is:

- (A) Pioneering radio-wave (wireless) communication research and improving the coherer around 1895
- (B) Building India's first national highway in 1853
- (C) Designing the Golden Quadrilateral road network
- (D) Inventing the twenty-foot shipping container (TEU)

42. Energy infrastructure includes dams, solar parks and wind farms. Sorting these by source, the correct statement is:

- (A) All three can supply renewable energy — water for the dam, sunlight for the solar park, and wind for the wind farm
- (B) All three burn coal to produce their energy
- (C) Only the dam is energy infrastructure; the others are social
- (D) None of them is infrastructure because nature provides the energy free

43. A 'utility' (from Latin utilis, 'useful') and a 'social-infrastructure service' are sometimes confused. Which item is a UTILITY rather than a social service?

(A) A government school

(B) A public hospital

(C) A district court

(D) Piped water supply to homes

44. The chapter argues energy and utilities matter to ALL other infrastructure. The reasoning behind this is that:

(A) Energy and utilities are used only inside private homes

(B) Transport, communication and social services all need power and water to function, so energy and utilities underpin them

(C) Energy and utilities replace the need for roads and schools

(D) Energy and utilities are unrelated to any other infrastructure

45. A city replaces a coal power plant with a large solar park. In the chapter's terms, the change makes the city's energy infrastructure more:

(A) Polluting, because solar parks burn more fuel than coal

(B) Sustainable, because it draws on a cleaner, renewable source with less environmental harm

(C) Social rather than physical infrastructure

(D) Useless, because renewable energy cannot power a city

46. Which statement about the relationship between energy infrastructure and a hospital is correct?

(A) The hospital is energy infrastructure because it consumes electricity

(B) Energy infrastructure is social because hospitals use it

(C) The two are unrelated, since hospitals never need power

(D) The hospital is social infrastructure that depends on energy infrastructure to keep its equipment and lights running

47. Schools, hospitals, courts and parks share a common feature in the chapter. That feature is that all four:

(A) Are social infrastructure aimed at people's wellbeing, learning, justice and recreation

(B) Are physical transport infrastructure that move goods

(C) Generate the nation's electricity

(D) Carry wireless communication signals

48. The chapter links social infrastructure to development through people. The most complete version of this link is:

(A) Social infrastructure speeds up cargo trains, raising exports

(B) Social infrastructure replaces the need for any physical infrastructure

(C) Schools and hospitals produce educated and healthy citizens, who can then work, invent and contribute to the country's growth

(D) Social infrastructure benefits only the people who build it

49. Sanitation appears under social infrastructure. The strongest reason it is grouped there rather than ignored is that:

- (A) Clean sanitation protects people's health and wellbeing, which is a core aim of social infrastructure
- (B) Sanitation makes aircraft fly faster
- (C) Sanitation increases a port's container count
- (D) Sanitation lengthens national highways

50. Two students argue about a vandalised public bus shelter. Student 1: 'Only the government must fix and protect it.' Student 2: 'Citizens share that duty too.' The chapter supports:

- (A) Student 1, because citizens have no role in public property
- (B) Student 2, because caring for public infrastructure is a collective responsibility shared by citizens and government
- (C) Neither, because public infrastructure needs no care at all
- (D) Student 1, because only foreigners maintain Indian infrastructure

51. Which set of citizen actions best demonstrates collective responsibility for shared infrastructure?

- (A) Breaking taps in a public washroom when no one is watching
- (B) Ignoring a leaking public pipeline because it is 'not my job'
- (C) Using a public park properly, not damaging benches, and reporting a broken streetlight
- (D) Waiting for a foreign agency to repair the local road

52. Why does the chapter say well-cared-for infrastructure 'serves everyone better'?

- (A) Damaged infrastructure actually helps a country develop faster
- (B) Only wealthy citizens ever benefit from public infrastructure
- (C) Maintained roads, parks and services last longer and stay usable, so the whole community keeps benefiting
- (D) Care makes no difference to how long infrastructure lasts

53. A village treats its hand-pump, school and approach road as 'everyone's to look after'. The civic principle this reflects is:

- (A) Private ownership — each item belongs to one family
- (B) Collective responsibility — shared public infrastructure is cared for jointly by the community
- (C) Government monopoly — citizens are barred from touching it
- (D) Foreign maintenance — outsiders must care for all of it

54. Sustainable infrastructure, per the chapter, is built so as to:

- (A) Use the cheapest, most polluting materials to save money
- (B) Ignore the environment so it can be built fastest

- (C) Use clean energy and eco-friendly materials, minimising pollution and harm to biodiversity
- (D) Rely only on ancient hand tools regardless of need

55. A living root bridge in Meghalaya and a concrete highway flyover are both infrastructure. The living root bridge best illustrates that infrastructure can be:

- (A) Only ever made of concrete and steel
- (B) Always harmful to the surrounding environment
- (C) Impossible to build without modern machines
- (D) Sustainable and in harmony with nature, grown by guiding tree roots across a stream

56. The living root bridges are made by the Khasi and Jaintia tribes of Meghalaya. The most important point the chapter draws from this example is that:

- (A) Infrastructure can only be modern and machine-built to last
- (B) Communities have long built durable infrastructure in harmony with nature using living materials
- (C) Natural materials can never form real infrastructure
- (D) Only the central government can build any kind of bridge

57. Kautilya's Arthashastra is cited in the chapter. Its mention is meant to show that:

- (A) Infrastructure was completely unknown until British rule
- (B) Only Western thinkers ever wrote about public works
- (C) Concern for roads and public works is an ancient Indian idea, not only a modern one
- (D) The Arthashastra was written after the Golden Quadrilateral was built

58. A student writes a one-line summary: 'Infrastructure is only visible roads and buildings, the British built the 1853 railways to help Indians, and caring for it is the government's job alone.' How many of the three claims are WRONG?

- (A) Only one is wrong
- (B) All three are wrong
- (C) Only two are wrong
- (D) None is wrong

59. A school principal lists facilities to label: (i) the school library, (ii) the road outside the gate, (iii) the electricity line powering the classrooms, (iv) the local court. How many are SOCIAL infrastructure?

- (A) One
- (B) Two
- (C) Three
- (D) Four

60. Reviewing the whole chapter, which single statement is FULLY correct with no embedded error?

- (A) Infrastructure is only physical, and the British railways of 1853 were built to benefit Indian travellers
- (B) Infrastructure means only invisible systems, and its upkeep is solely the government's responsibility
- (C) Infrastructure includes social services, but J.C. Bose built India's first railway in 1853
- (D) Infrastructure, the engine of development, has physical and social arms and is best kept working through care shared between government and citizens

Solutions — Social Science (Class 7), Chapter 19: Infrastructure: Engine of India's Development — Paper 3

Paper 3 — (progressively harder)

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	11	B	21	C	31	A	41	A	51	C
2	C	12	B	22	D	32	A	42	A	52	C
3	C	13	C	23	A	33	C	43	D	53	B
4	A	14	D	24	D	34	A	44	B	54	C
5	A	15	D	25	B	35	D	45	B	55	D
6	C	16	B	26	A	36	A	46	D	56	B
7	D	17	D	27	A	37	D	47	A	57	C
8	D	18	A	28	B	38	D	48	C	58	B
9	C	19	C	29	D	39	B	49	A	59	B
10	C	20	B	30	A	40	B	50	B	60	D

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (C) Physical infrastructure covers transport, utilities, communication and energy. The fibre-optic network (communication), the water pipeline (utility) and the wind farm (energy) are all physical, giving three. The district court is a social-infrastructure facility, so it is not counted. Answers of two or one undercount the physical items, and 'all four' wrongly treats the court as physical rather than social.

2. (C) Infra means 'below' and structura means 'a building/structure', so the literal sense is the underlying base that lies beneath and supports the rest. 'Tallest structure above' inverts infra into 'above'. 'Built quickly and cheaply' confuses it with a speed term. 'Shared between far-apart places' borrows the tele/communicare roots of telecommunication, not infrastructure.

3. (C) The claim fails because much essential physical infrastructure is invisible — buried power grids, water pipelines and communication networks. They are unseen yet vital,

directly contradicting 'cannot be seen, cannot be infrastructure'. The hospital, highway and port are all highly visible, so they cannot rebut a claim that is specifically about invisible systems.

4. (A) The chapter's two types are physical and social — a functional division (systems vs services for people), not a price one. 'Three official types' contradicts the two-fold split. 'All costs the same' is false. 'Cheap never counts' is invented. Only the function-based reasoning correctly explains why cost is the wrong axis.

5. (A) The chapter calls infrastructure the engine (driver) of development and divides it into physical and social arms — both ideas appear together only in the correct option. Limiting it to 'roads and buildings', or to 'only social services', or to 'only invisible systems', each repeats a misconception the chapter rejects, and 'enemy/final result' misstates its role.

6. (C) Sanitation supply systems are physical utilities; the chapter does list sanitation as a social-infrastructure aim too, but a treatment PLANT and pipes are the physical-utility hardware. A public park is a social-infrastructure facility. A mobile-tower network is physical communication. Only the chosen set assigns each correctly; the others either lump all as social, or swap the park and the network between physical and social.

7. (D) The chapter stresses that physical and social infrastructure are two arms working together; strong physical systems without healthy, educated people leaves development incomplete. 'Physical alone guarantees progress' and 'social plays no role' both deny the social arm. Demolishing working physical infrastructure would harm, not help, development.

8. (D) Infrastructure is the set of basic facilities a country needs; it includes BOTH social services like hospitals and physical systems like power grids, which is why both qualify. 'Both physical because both use electricity' misclassifies the hospital. 'Both social' misclassifies the grid. 'Only roads count' is the very misconception the chapter rejects.

9. (C) Power supply is physical energy/utility infrastructure, and because transport, communication and social services all need power, it underpins the rest. Calling it social confuses 'serves people' with the physical/social divide. It carries energy, not communication signals. Private household use does not strip a public supply system of being infrastructure.

10. (C) A court is social infrastructure (a service institution), so it is the odd one out. A gas pipeline is a physical utility, a telecom uplink is physical communication, and a hydroelectric dam is physical energy infrastructure — all three are physical.

11. (B) Good physical infrastructure (power, roads) plus social infrastructure (schools) gives X the foundation for economic and human development — the backbone idea. 'Y

faster with less to maintain' and 'too much slows X' both invert the chapter's claim, and 'infrastructure does not affect development' contradicts its central thesis.

12. (B) The backbone metaphor highlights support and connection: economic activity rests on and is linked by transport, power, water and communication. The metaphor is not about expense, nor about rigidity/permanence, nor about serving only 'large organs'; reading it that way mistakes a structural-support image for cost, inflexibility, or selectivity.

13. (C) Transport infrastructure moves inputs and outputs quickly while energy infrastructure powers the machines, lowering cost and time — a multi-link cause-effect the chapter supports. The other options either claim infrastructure raises a factory's costs, reduce its benefit to cutting headcount, or deny any effect, all contrary to the backbone-of-the-economy idea.

14. (D) Physical systems handle the flow of goods, energy and information, while social services build human capability — the two complementary roles. The first distractor swaps the two roles. 'Both only move goods' erases the human-development role. 'City-only vs village-only' invents a geographic split the chapter does not make.

15. (D) Road-network size and railway passenger numbers are distinct facts; the chapter gives 'second-largest road network' and 'about 20 million rail passengers a day' separately, without claiming a matching rank. 'Always the same' is an invalid leap, 'no railways at all' contradicts the chapter, and 'smallest roads' denies the stated fact.

16. (B) An expressway is a high-speed inter-state national road built by the central government — that is its defining contrast with a village lane. A village-maintained local road is the opposite of an expressway. An elevated urban rail line describes a metro, and a coastal cargo route describes sea/port transport, neither of which is a road expressway.

17. (D) 'Quadrilateral' means four-sided; the network joins four major cities (Delhi, Mumbai, Chennai, Kolkata) into a four-sided circuit. It is not a village project, the name does not refer to lane count, and the four cities are Indian, not foreign capitals.

18. (A) NH44 is a major (notably long) national highway connecting far-apart regions. The other options describe entirely different infrastructure types — a metro (urban rail), a port (sea transport), and a solar park (energy) — none of which is a national highway.

19. (C) The Golden Quadrilateral is transport infrastructure; its benefit to a goods company is rapid movement between major cities. Supplying electricity is energy infrastructure, sewage treatment is a utility/social function, and a patent is a legal right over an invention — none of these is what a highway network provides.

20. (B) Roads, rail, metros, airports and ports all carry people or goods, so they are forms of physical transport infrastructure. They are not social services, not communication systems (which carry information), and not energy systems (which carry power).

21. (C) The chapter's key point is a change of purpose: built in 1853 for colonial extraction and control, the railways became India's lifeline only afterward. Saying they served Indians 'from the first day' keeps the colonial-benefit misconception; 'never a lifeline / still British' denies their present role; and 'built by Indians for free travel in 1853' is historically false.

22. (D) The chapter lists export of raw materials, sale of British goods, and rapid troop movement as the real colonial motives — so those three are NOT exceptions. The one that does NOT belong is 'free fast travel for villagers' benefit', which is precisely the myth the chapter corrects.

23. (A) Twenty million divided by twenty equals one million, so reading '20 million' as '20' understates the figure by roughly a factor of a million. Factors of a hundred, a thousand, or ten are all far too small to span the gap between 20 and 20,000,000.

24. (D) Railways built for colonial gain became a national lifeline, showing infrastructure can be repurposed for new and broader ends. 'Must be torn down', 'only one purpose', and 'useless within years' each contradict this lesson of adaptive, lasting value.

25. (B) Metros are intra-city rapid transit reducing road congestion, whereas the Indian Railways span the whole country between regions. The first distractor swaps their scopes. Both are rail, not sea transport. And both are physical transport, not energy or communication infrastructure.

26. (A) A metro's specific value is mass urban commuting that takes pressure off congested roads. Exporting raw materials describes colonial-era freight rail/ports, generating electricity is energy infrastructure, and moving containers between coasts is sea/port transport — none is the metro's role.

27. (A) Avis ('bird') points to flight, so aviation infrastructure is airports and aircraft moving through the air. Harbours and ships are sea transport, power stations are energy, and water pipelines are utilities — none matches the 'bird'/air meaning.

28. (B) The figure 159 refers to airports; sea ports are different facilities located on the coasts, and the chapter does not equate the two counts. 'Always equal' is an unwarranted leap, 'no airports' contradicts the figure, and 'all 159 are sea ports' misattributes the number.

29. (D) Airports (air) and sea ports (water) are both modes of moving people and goods, hence both physical transport infrastructure. They are not social services, not energy systems, and not communication networks.

30. (A) TEU (Twenty-foot Equivalent Unit) counts standard shipping containers, so $8,000 - 5,000 = 3,000$ more containers at Port B, showing larger throughput. TEU is not a measure of train distance, of height, or of aircraft, so the other readings misuse the unit.

31. (A) A standard equivalent unit lets very different ports be compared on the same scale of container throughput. TEU has nothing to do with metro speed, crane height, or airport ownership, so those options misidentify what the unit standardises.

32. (A) Having ports on both coasts opens trade routes in two directions, widening import-export access. Ports are not useless for trade, mountains like the Himalayas have no sea access for ports, and coastal ports do not multiply airports — those options confuse port geography with other facts.

33. (C) Container shipping uses a sea port; fast air freight uses an airport — both physical transport infrastructure suited to each task. Power grids/metros are not for overseas cargo or parcels, schools/hospitals are social services, and pipelines/courts are utilities and social facilities, none of which ships goods abroad.

34. (A) J.C. Bose's work on radio-wave communication and the coherer is the chapter's evidence that India contributed to communication science, rebutting the 'entirely Western' claim. 'Only after 1947' and 'ancient trade use' are unsupported, and tying radio to the 1853 railways confuses two unrelated facts.

35. (D) The railways came in 1853, Bose's wireless work about 1895, and independence in 1947 — so $1853 < 1895 < 1947$ is the right order. The other sequences place 1895 before 1853, or 1947 before 1853, or Bose's 1895 work after 1947, all of which violate the actual dates.

36. (A) The coherer was an early detector that picked up and decoded wireless signals, which Bose improved. It did not generate railway power, pump water (a utility function), or carry passengers (a transport function), so those options confuse communication with energy, utilities, and transport.

37. (D) Choosing not to patent meant Bose did not lock up his ideas for personal exclusive gain, leaving them open for others' use. It does not imply the work was worthless, that he was legally barred, or that he kept everything secret — declining a patent is the opposite of hiding work away.

38. (D) A patent grants the holder the sole right to make, sell or use the invention for a set period, so patenting would have legally restricted others for those years. Patenting does not affect whether devices function, has nothing to do with ending colonial rule, and 'everyone could copy freely' is the opposite of what a patent does.

39. (B) Tele ('far') + communicate ('to share') means sharing information over distance, which a long-distance phone call exactly does. Driving on a highway is transport, loading containers is sea transport/logistics, and generating power is energy — none of these is sharing information at a distance.

40. (B) The essential split is what each moves: communication carries information, transport carries people and goods. The first distractor swaps these roles, 'both move only electricity' describes energy not these two, and 'communication abroad only / transport India only' invents a false geographic divide.

41. (A) Bose is credited with early radio-wave research and improving the coherer about 1895. He did not build highways (and 1853 is the railway date), did not design the Golden Quadrilateral, and did not invent the shipping container — those mix him up with unrelated transport facts.

42. (A) Hydro (dams), solar parks and wind farms all draw on renewable natural sources — water, sun and wind. None of them burns coal, all three are energy infrastructure (not social), and using a free natural source does not stop the built systems from being infrastructure.

43. (D) A utility is a basic supplied service like water, electricity or gas — piped water supply fits exactly. A school, a hospital and a court are social-infrastructure institutions, not utilities, so they are the wrong choices for a 'utility'.

44. (B) Trains, phones, hospitals and schools all depend on power and water, so energy and utilities form a base layer beneath the rest. They are not home-only, they support rather than replace roads and schools, and they are far from unrelated to other infrastructure.

45. (B) Switching to solar uses a clean, renewable source, reducing pollution — the definition of more sustainable infrastructure. Solar does not burn fuel (so it is not more polluting), it remains physical energy infrastructure (not social), and renewables are not useless for cities.

46. (D) A hospital is a social-infrastructure service, but it relies on energy infrastructure to operate — showing how the layers depend on each other. Consuming electricity does not make a hospital energy infrastructure, nor does a hospital's use make a power grid social, and hospitals certainly do need power.

47. (A) Schools, hospitals, courts and parks are all social infrastructure serving human needs — health, education, justice and recreation. None of them moves goods (transport), generates electricity (energy), or carries signals (communication), so the other options misclassify the whole group.

48. (C) Social infrastructure develops human capability — health and education — which feeds back into national growth. It does not speed cargo (that is transport), cannot replace physical infrastructure, and benefits society broadly, not just its builders.

49. (A) Sanitation safeguards health and wellbeing, the central purpose of social infrastructure, so it belongs there. It has no bearing on aircraft speed, port throughput, or

highway length, so those options confuse a health measure with unrelated transport facts.

50. (B) The chapter teaches that maintaining public infrastructure is a collective duty — citizens as well as the government — so Student 2 is right. Student 1's 'government-only' view is the very misconception corrected, 'needs no care' is false, and 'only foreigners maintain it' is unsupported.

51. (C) Proper use, avoiding damage, and reporting faults are exactly the citizen behaviours the chapter calls collective responsibility. Breaking taps and ignoring a leak harm shared infrastructure, and waiting on a foreign agency abdicates the shared duty — all three are the opposite of responsible care.

52. (C) Care preserves infrastructure so it endures and remains useful to all, benefiting the community over time. Damage does not aid development, public infrastructure is for everyone not only the wealthy, and care clearly does affect lifespan — so the distractors each contradict the chapter.

53. (B) Treating shared facilities as everyone's to maintain is collective responsibility. It is not private family ownership, not a government monopoly that bars citizens, and not reliance on outsiders — those misread the shared-duty idea.

54. (C) Sustainable infrastructure deliberately uses clean energy and eco-friendly materials to limit pollution and protect biodiversity. The other options describe the opposite — pollution to cut cost, ignoring the environment for speed, or an irrelevant tool restriction.

55. (D) Living root bridges, grown by the Khasi and Jaintia tribes by training tree roots across streams, show infrastructure that is sustainable and nature-friendly. They disprove the ideas that infrastructure must be concrete-and-steel, must harm nature, or must need modern machines.

56. (B) The example shows lasting, nature-friendly infrastructure built with living materials by local communities. It contradicts the notions that durable infrastructure must be machine-built, that natural materials cannot count, or that only the central government can build bridges.

57. (C) The Arthashastra's discussion of public works shows that caring for infrastructure is an ancient Indian concern, giving historical depth to the topic. It was not unknown before the British, not a solely Western idea, and the Arthashastra long predates the modern Golden Quadrilateral.

58. (B) Each claim is a misconception the chapter corrects: infrastructure also includes social services and invisible systems (so 'only visible roads and buildings' is wrong); the 1853 railways served British interests, not Indians' benefit (wrong); and care is a collective

duty of citizens too, not the government's alone (wrong). All three being wrong, 'one', 'two', or 'none' all undercount the errors.

59. (B) Social infrastructure covers institutions serving people's wellbeing, learning and justice. The library (part of a school) and the court are social, giving two. The road is physical transport and the electricity line is physical energy, so they are not counted. 'One' misses one social item, while 'three' or 'four' wrongly count the road and/or power line as social.

60. (D) The correct statement combines several chapter ideas accurately: infrastructure drives development, splits into physical and social arms, and is sustained by shared care. Each distractor hides an error — denying the social arm and the true colonial railway motive, restricting infrastructure to invisible systems with government-only care, or falsely crediting J.C. Bose (a communication scientist, c.1895) with building the 1853 railway.